



# Effect of rice husk ash-derived activator on the structural build-up of alkali activated materials

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## ABSTRACT

Waste-derived sodium silicates (SSs) were synthesised from rice husk ash (RHA) to produce alkali activated materials (AAMs) using fly ash and slag as precursors. The effects of these SSs on the structural build-up were investigated by monitoring the evolution of the viscoelasticity, ultrasonic pulse waves, setting time, and infrared spectra. Two types of RHA derived SSs (filtered SS and unfiltered SS) were prepared to investigate the effect of the undissolved residue on the hardening process. The derived SS contained a large portion of monomer silicate species. The results showed that the structural build-up mechanism was comparable between the AAMs activated by the filtered SS and a commercial SS. The unfiltered SS increased the yield stress of the paste by up to 10 times. However, the residues in the unfiltered SS had insignificant impacts on the gel formation and compressive strength when using identical  $\text{SiO}_2/\text{Na}_2\text{O}$  and  $\text{Na}_2\text{O}/\text{binder}$  ratios in the mix design.

## 1. Introduction

### 1.1. Background

The significant upcoming challenge in the construction industry and engineering is constructing and restoring infrastructure in an economical, durable, and sustainable manner while reducing the environmental impact. Several pathways have been pursued to reduce the environmental impacts of the construction materials and the dependency on Portland cement as a standalone binder [1–5]. One pathway focuses on the development of low-carbon cementing binder systems [6]. While introducing various types of low-carbon embedded cementitious systems, alkali-activated materials (AAMs) have emerged as a means of mitigating  $\text{CO}_2$  emissions from the construction sector [7]. Alkali activation is the process in which an alkaline medium is used to transform aluminosilicate sources (precursors) into synthetic aluminosilicate gels (when using low-calcium precursor) or aluminium-modified calcium silicate hydrate gels (when using calcium-rich precursor) [8,9]. Alkaline hydroxides and silicates are the major industrial activators used in AAMs. However, their manufacturing process is energy-intensive that leads to high economic and environmental impacts [10]. For example, the production of sodium silicate activator generates major  $\text{CO}_2$  emissions through the decomposition and calcination processes of sodium carbonate ( $\text{Na}_2\text{CO}_3$ ) with silica sand [11]. Despite this fact, previous

assessments on AAMs concluded that the greenhouse gas emission of AAM concrete is 32%–64% lower than the one from Portland cement concrete of similar strength [1,12,13]. It is interesting to note that alkaline hydroxides and silicates were responsible for ~60% of the total environmental impacts of AAMs [14]. A comparative analysis on AAMs indicated that the environmental impact could be almost halved by using waste-derived sodium silicates compared to AAMs activated using commercial sodium silicate [15]. This indicates that the sustainable supply of waste-derived activators can substitute the commercially available alkali silicate activators and, consequently, promote the sustainability of AAMs.

Rice husk ash (RHA) resulting from the combustion of rice husk in biomass power plants can be a promising source of silica for AAM activators. 770 million tonnes of paddy rice are annually produced worldwide [16], and consequently, the annual production amount of RHA is around 26–34 million tonnes [17]. RHA contains about 90% of completely or largely amorphous silica [18] that could be potentially incorporated in the synthesis of alternative sodium silicate activators [1]. To obtain an alternative activator using silica-rich resources, silica needs to be extracted at high temperatures within an alkaline medium, the so-called ‘hydrothermal method’ [19–21]. This process involves the dissolution of RHA into an alkaline solution at temperatures up to 100 °C [20,22,23]. Although researchers have yet to develop the best-optimised extraction process, the yield of silica was found to be in the range

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between 12% and 95%, depending on the processing temperature and processing time [20,24,25]. The silicate species in the activator is a determinant component in the polycondensation reaction (i.e., geopolymerization), which in turn affects the overall performance of AAM concrete [26]. For example, in comparison with AAMs made of commercial activator, some studies reported similar compressive strengths for AAMs using the activators derived from silica-rich waste resources [15,20,25,27,28], while the strength was lower in other studies [24,29,30]. This variation could be the result of the variation of available silicate species in the activators derived from silica-rich waste resources.

## 1.2. Research significance

Numerous studies on the RHA-derived activators have been mainly focused on the mechanical and microstructural characteristics of extracted activators [24,31,32]. However, the silicates in the activators also significantly affect the temporal structural build-up and setting of AAMs because the composition of the activator plays a pivotal role in the dissolution kinetics of the precursor at the early age of the reaction process [33–35]. To the best knowledge of the authors, there are no studies to date on the rheological behaviour of AAMs incorporated with the RHA-derived activators. Previous studies by the authors [34,35] showed that the rheological properties of AAMs are distinctively different from those observed in Portland cement-based systems. Therefore, to promote the practical viability of not only AAM but also the waste-derived activators, it is required to understand the effects of the RHA-derived activator on rheological properties, including yield stress (the critical threshold stress for flowing), the plastic viscosity, and the structural build-up due to the chemical reaction in the AAMs.

This paper aims to investigate the effect of RHA-derived sodium silicate (SS) solution on the time-dependent properties of AAM pastes from the fresh condition to the final setting. The RHA-derived SS was synthesised using a hydrothermal process [20]. Two types of RHA-derived SS were prepared. One was filtered to obtain the clean SS without residues, and the other one was used without filtration to investigate the effect of the undissolved particles on the rheological properties of AAM pastes. Flow sweep and stress growth tests were conducted at elapsed resting times to determine the change in dynamic and static yield stress of the AAM pastes. The evolution of the viscoelastic parameters, namely, shear moduli (storage modulus and loss modulus) and phase angle within the linear viscoelastic domain (LVED), were examined using the small amplitude oscillatory shear (SAOS) test in order to study the effect of the synthesised solution on the percolation process of the particles. The SAOS tests were well evaluated as one of the techniques to investigate the temporal viscoelastic behaviour of cementitious systems, including AAMs [35–37]. These measurements were then compared with an in-situ Fourier transform infrared (FTIR) spectroscopy to correlate between the evolution of the interparticle interactions and the transformation of the solid precursor into a synthetic geopolymeric gel [38]. The velocity evolution of ultrasonic pulse waves (P-waves) was monitored to investigate the setting process and to compare with the change in the rheological properties. This transmission time of ultrasonic waves can be correlated with the evolution of the storage modulus obtained from rheological measurements [39–41]. Finally, the impact of the RHA-derived sodium silicate on the compressive strength of AAM pastes was also examined.

## 2. Experimental program

### 2.1. Materials

Siliceous fly ash (FA) (classified as Class F fly ash according to ASTM C618 [42]) and ground granulated blast furnace slag (GGBFS) from Blue Circle Southern Cement Australia, were used as precursors. RHA was purchased from NK Enterprises, India. The chemical compositions of

RHA, FA and GGBFS, obtained from wavelength-dispersive X-ray fluorescence (PANalytical AXIOS instrument), can be found in Table 1. The specific gravities of RHA, FA and GGBFS were 2.2, 2.1 and 2.8, respectively. The particle size distribution was also determined using the Malvern Mastersizer based on laser diffraction and is presented in Fig. 1. The median particle size of RHA, FA and GGBFS was 18  $\mu\text{m}$ , 22  $\mu\text{m}$  and 11  $\mu\text{m}$ , respectively.

A mixture of sodium hydroxide pellets (NaOH) and sodium silicate (SS) solutions (either commercial or RHA-derived SS) was used as an alkaline activator. NaOH (98.0% purity) was supplied by Chem-Supply Pty Ltd. Australia. The commercial SS solution supplied by PQ Australia had the following composition: 14.7% of  $\text{Na}_2\text{O}$ , 29.4% of  $\text{SiO}_2$ , 55.9% of water (by weight). The RHA-derived SS was prepared using the hydrothermal method described in the following section.

### 2.2. Sodium silicate synthesis from RHA

A hydrothermal process was carried out to dissolve RHA and NaOH pellets in water. The proportions of the suspension mix consisted of 486 g of RHA, 288 g of NaOH pellets and 785 g of water. These mix proportions allowed for the theoretical  $\text{SiO}_2/\text{Na}_2\text{O}$  weight ratio of 2.0 that is equivalent to the ratio in the commercial SS used, considering 91.9 wt%  $\text{SiO}_2$  in RHA (see Table 1) and the 77.5 wt%  $\text{Na}_2\text{O}$  in NaOH. NaOH pellets and water were first mixed and allowed to cool down to room temperature. After that, RHA was added into the glass beaker containing prepared NaOH solution, and then the glass beaker was placed on a temperature-controlled hot plate. The heating process started, consisting of continuous stirring at 80 °C for 1 h using a magnetic stirrer. Previous studies showed that 80 °C and 1 h were sufficient to dissolve silica from RHA [20,25]. The prepared solution was then filtered using Grade GF/C glass microfibre filters (1.2  $\mu\text{m}$  pore size) to remove the undissolved particles. The unfiltered solution was also prepared separately to study the effect of the residues on the AAM paste performance. The filtered/unfiltered solutions were then stored in high-density polyethylene bottles until their usage in the AAM pastes. The solid residue after the filtration was washed with isopropanol to remove any traces of the solution, dried in the oven to remove the free water, and then stored in the vacuum desiccator.

The filtered solution was analysed using the inductively coupled plasma-optical emission spectrometer (Agilent 720-ES ICP-OES) at Solid State & Elemental Analysis Unit at Mark Wainwright Analytical Centre in the University of New South Wales (UNSW). Due to the strong base media of the solution, it was diluted with deionised water to measure Si and Na concentrations. The conventional 5% nitric acid digestion and dilution was not used as it resulted in precipitation in the solution [43].

The solid residues were analysed using X-ray diffraction (XRD), FTIR and thermogravimetric analysis (TGA) to quantify the amounts of undissolved silica, unburnt carbon and other oxides. In XRD analysis,  $\text{Cu-K}\alpha$  radiation with a wavelength of 0.15418 nm operating at 45 kV and 40 mA and 0.013° 2 $\theta$  step size with a sample spinner were used. Rietveld XRD refinement was carried out to quantify the amorphous and crystalline phases in RHA by spiking the RHA with 20 wt% of pure silicon as an internal standard. TGA was conducted using the Q600 SDT from TA

**Table 1**  
Chemical compositions of RHA, fly ash and GGBFS.

| Chemical compositions (%) | RHA   | Fly ash | GGBFS |
|---------------------------|-------|---------|-------|
| $\text{SiO}_2$            | 91.86 | 66.56   | 31.52 |
| $\text{Al}_2\text{O}_3$   | 0.61  | 22.47   | 12.22 |
| CaO                       | 1.03  | 1.64    | 44.53 |
| $\text{Fe}_2\text{O}_3$   | 0.58  | 3.54    | 1.14  |
| MgO                       | 0.67  | 0.65    | 4.62  |
| $\text{Na}_2\text{O}$     | 0.14  | 0.58    | 0.21  |
| $\text{K}_2\text{O}$      | 1.91  | 1.75    | 0.33  |
| $\text{SO}_3$             | 0.06  | 0.10    | 3.24  |
| LOI                       | 2.00  | 1.66    | 0.79  |

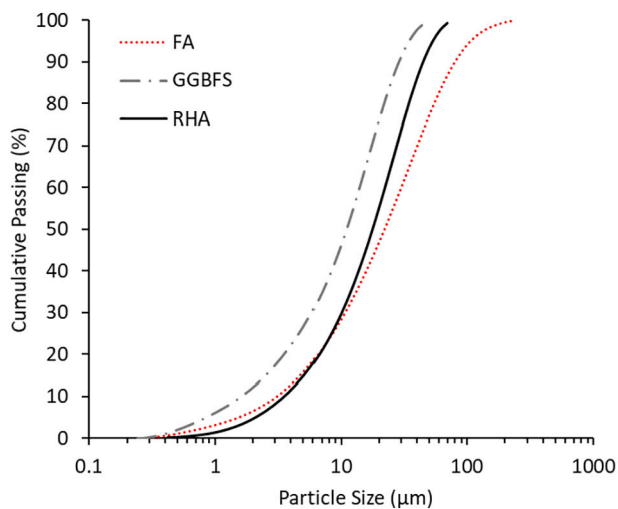


Fig. 1. The particle size distributions of RHA, FA and GGBFS.

Instrument. Due to the minimal quantity of the residue used in TGA (~20 mg), a representative sample (5–10 g) was collected from a larger volume of powders using the sample splitter. Then, ~20 mg of the residual powder was heated in an alumina crucible up to a temperature of 1050 °C with a heating rate of 10 °C/min with a nitrogen purge of 50 ml/min. Five minutes of the isothermal period at 100 °C was applied, and the weight (%) was normalised based on the weight of the sample at 100 °C.

FTIR spectra were collected from three liquid activators (commercial SS, unfiltered RHA SS, and filtered RHA SS) and two solids (the raw RHA powder and the residue after the extraction process) to study the main bands assigned to the Si—O—Si bonds. The measurements were conducted using a Perkin Elmer Frontier IR spectrophotometer by placing a small portion of the material on a 3 mm diameter diamond/ZnSe crystal of ATR. The spectra were then collected at the IR range from 700 to 1800  $\text{cm}^{-1}$ . The range of wavenumber was selected because Si—O—Si bonds in the raw RHA and the undissolved particles are observed in the spectrum between 700 and 1300  $\text{cm}^{-1}$  and the water band is at 1643  $\text{cm}^{-1}$  [44]. The spectral resolution was 4  $\text{cm}^{-1}$  and an average of four scans was taken for each sample.

### 2.3. Mix design and mixing procedure

Three different pastes were prepared, as shown in Table 2. The  $\text{SiO}_2/\text{Na}_2\text{O}$  weight ratio in the activator and  $\text{Na}_2\text{O}/(\text{FA} + \text{GGBFS})$  were kept constant at 1.0 and 7.5%, respectively. GGBFS content was fixed at 20%

**Table 2**  
Mix proportions of AAM pastes.

| Mix                                              | CSS  | RSS-F | RSS-UF |
|--------------------------------------------------|------|-------|--------|
| GGBFS ( $\text{kg}/\text{m}^3$ )                 | 295  | 295   | 295    |
| Fly ash ( $\text{kg}/\text{m}^3$ )               | 885  | 885   | 885    |
| % GGBFS (vol.)                                   | 20%  | 20%   | 20%    |
| NaOH 10 M ( $\text{kg}/\text{m}^3$ )             | 185  | 140   | 140    |
| Commercial SS ( $\text{kg}/\text{m}^3$ )         | 300  | 0     | 0      |
| RHA-derived SS ( $\text{kg}/\text{m}^3$ )        |      |       |        |
| Filtered                                         | 0    | 343   | 0      |
| Unfiltered                                       | 0    | 0     | 370*   |
| Add. water ( $\text{kg}/\text{m}^3$ )            | 135  | 135   | 135    |
| Liquid/binder (vol.)                             | 0.9  | 0.9   | 0.9    |
| w/s (wt.)                                        | 0.32 | 0.32  | 0.31   |
| $\text{SiO}_2$ ( $\text{kg}/\text{m}^3$ )        | 88   | 88    | 88     |
| $\text{Na}_2\text{O}$ ( $\text{kg}/\text{m}^3$ ) | 88   | 88    | 88     |
| $\text{SiO}_2/\text{Na}_2\text{O}$ (wt.)         | 1    | 1     | 1      |
| $\text{Na}_2\text{O}/\text{binder}$ (wt.)        | 7.5% | 7.5%  | 7.5%   |

\* This quantity includes 27  $\text{kg}/\text{m}^3$  residues.

by volume (25% by weight) based on trial mixes to provide suitable flowability and setting time. Detailed results of the trial mixes are not the scope of this paper but can be found in reference [34,45]. This mix proportion was also found to be suitable for calcium-rich AAMs, according to Soutsos et al. [46]. Two types of RHA SS solutions were used to investigate the effects of filtered (RSS-F mix), and unfiltered (RSS-UF mix) solutions on the behaviour of AAM pastes in comparison with the control AAM paste prepared with the commercial SS (CSS mix). As will be explained in Section 3.1, RHA was not completely dissolved during the hydrothermal process, and hence, the amounts of SS and NaOH in the mixes RSS-F and RSS-UF were adjusted based on the dissolved silica from RHA to keep the  $\text{SiO}_2/\text{Na}_2\text{O}$  and  $\text{Na}_2\text{O}/(\text{FA} + \text{GGBFS})$  ratios at 1.0 and 7.5%, respectively (see Table 2). In addition, when using the unfiltered RHA SS (RSS-UF), the undissolved particles were considered as a filler material.

In this study, the mixing of pastes was performed using the Renfert vacuum mixer to remove any effect of entrapped air bubbles in the pastes [47,48]. The activating solution, FA and GGBFS were mixed at 140 rpm for 30 s followed by 280 rpm for 60 s, in accordance with ASTM C305 [49]. Immediately after mixing (~5 min), the homogeneously mixed paste was used for all measurements. More details about the mixing process and the validation of this process for the rheological studies can be found in the previous literature [34,35].

### 2.4. Rheological measurements

Different rheological measurement tests were carried out on the AAM pastes at 25 °C using Discovery Hybrid Rheometer (DHR-2) from TA Instruments. A sandblasted concentric cylinder geometry was used. The inner radius of the cup cylinder was 15.16 mm, and the radius of the bob cylinder was 14.00 mm, making the gap width as 1.16 mm. The bob cylinder has a conical bottom with an apex angle of 120°. Immediately after mixing, 23 ml of the paste was poured into the cup cylinder followed by a pre-shear at 100  $\text{s}^{-1}$  for 60 s to achieve a well-dispersed state [50–53]. When the paste was tested at different elapsed times, a new sample was prepared for each measurement to avoid any effect of shearing history from the previous measurement.

#### 2.4.1. Flow sweep test for dynamic yield stress

Flow sweep tests were performed to determine the yield stress and the plastic viscosity of the AAM paste, as well as the Newtonian viscosity of the activators used in preparing the AAM pastes. To measure the Newtonian viscosity of the activator, a linear flow sweep was performed using 20 points between shear rates 0.1  $\text{s}^{-1}$  and 100  $\text{s}^{-1}$ . For AAM pastes, a logarithmic flow sweep was performed (using 10 points per decade and 10 s/point) at elapsed times of 5, 10, 20, 30 and 40 min.

Alnahhal et al. [34] found that the additional time needed for the descending curve can increase the potential effect of chemical changes on the rheological parameters. Therefore, the ascending shear rate-shear stress profile was chosen for analysis. The points between 0.1 and 10  $\text{s}^{-1}$  were found to fit well with the Bingham model. Thus, this range was chosen to determine the dynamic yield stress of the paste at each elapsed time. One of the drawbacks of the flow sweep test in determining the dynamic yield stress is that this test does not capture the required data (stress vs shear rate for a flow curve) at a specific moment in time. At each step (i.e. at each shear rate), the test took 10 s consisting of 5 s of equilibrium time and 5 s of averaging time in order to obtain the stable value of the shear stress. The 10 s of measurement at each step was determined by the results of the three interval thixotropy test. Thus, the effect of chemical changes that occur during the time required for the flow sweep test can affect the value of the dynamic yield stress. To study the yield stress for chemically active materials like AAMs, tests with short time measurement may be the better approach at each elapsed time. Therefore, the stress growth test was also conducted as explained in Section 2.4.2.



#### 2.4.2. Stress growth test for static yield stress

The static yield stress is measured at elapsed times of 5, 10, 20, 30 and 40 min by applying a constant shear rate of  $0.05 \text{ s}^{-1}$  (constantly increasing strain). When the paste is sheared at a constant small shear rate, the stress evolves and reaches a peak value or plateau, suggesting the beginning of the flow. This value is defined as the static yield stress. This static yield stress was compared with the dynamic yield stress determined by the flow sweep test.

#### 2.4.3. SAOS for viscoelastic measurements and structural build-up

The temporal changes in the viscoelastic behaviour of AAM paste can be monitored in a non-destructive regime by applying a small oscillation strain within the LVED. Therefore, the critical strain (the largest oscillatory strain within the LVED) was first determined by sweeping the oscillatory strain from 0.001% to 20% at a constant frequency of 1 Hz. The critical strain of the LVED corresponds to the point where the decrease in the storage modulus starts, as shown in Fig. 2. The critical strain for all mixes was found to be between 0.026% and 0.040%. Therefore, a shear strain of 0.01% was selected to perform the time sweep measurements. The evolution of the three essential viscoelastic parameters was examined in the SAOS test, namely, storage modulus ( $G'$ ), loss modulus ( $G''$ ) and phase angle ( $\delta$ ). The viscoelastic behaviour takes place between  $\delta$  of  $0^\circ$  and  $90^\circ$ , with  $0^\circ$  corresponding to the rigid behaviour and  $90^\circ$  corresponding to the liquid behaviour [54].

#### 2.5. Setting time and ultrasonic pulse velocity test

An automatic Vicat needle apparatus (VICAMATIC-2) was used to determine the initial and final setting times in accordance with ASTM C191 [55]. The initial setting time was determined when the penetration depth reached 25 mm.

The test procedure for P-wave velocity is outlined in ASTM C597 [56]. The test was performed using a Pundit PL-200 instrument with 54 kHz single-crystal transducers. In this paper, the test method consisted of collecting the transmission information of P-waves through the paste immediately after mixing for 24 h. This test could assess the changes in the ultrasonic signal transmission due to microstructural changes (mainly hardening process), which were correlated to the changes in the viscoelastic behaviour over time. P-waves were sent every 1 min through a 20 mm-wide paste sample, and the data was collected over 24 h. The test configuration is shown in Fig. 3. A fresh AAM paste was poured in a 20 mm-wide U-shaped foam rubber container. Foam rubber has low wave-transmittance, and hence, it allows the P-wave signal to travel through the paste from the transmitter to the receiver without being distorted [57]. A thin film was placed between the transducers and the paste to avoid damage in the transducers by the alkaline activator.

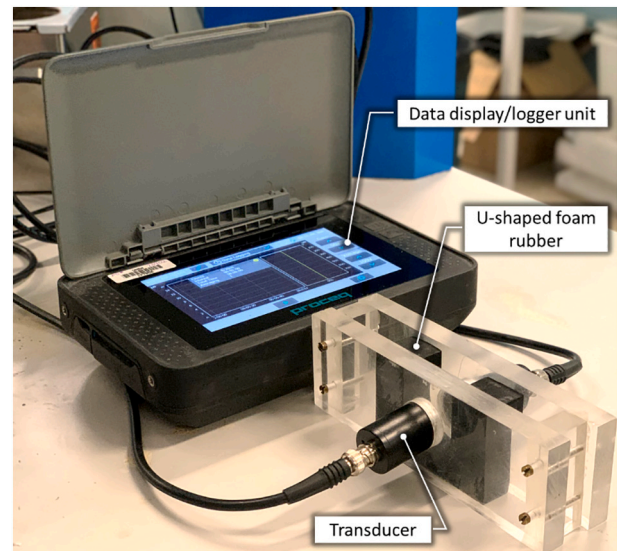


Fig. 3. Ultrasonic pulse velocity test configuration.

#### 2.6. FTIR for the evolution of gel formation

The attenuated total reflectance (ATR)-FTIR method was carried out to study the gel formation mechanism of the AAM pastes prepared with commercial and RHA SS. AAM pastes were mixed using the method prescribed in Section 2.3, and part of the paste was poured into a polyethylene reaction cell. After that, the set was mounted onto a 3 mm diameter diamond/ZnSe crystal of ATR and sealed to avoid any loss of water. The spectra were then collected at 1 min intervals over 2 h at room temperature ( $23 \pm 2^\circ \text{C}$ ). The absorbance intensity was measured at the mid IR range from  $700 \text{ cm}^{-1}$  to  $1800 \text{ cm}^{-1}$  because the change in the Si—O—Si bonds due to the reaction products can be usually observed between  $900 \text{ cm}^{-1}$  and  $1000 \text{ cm}^{-1}$  [58].

#### 2.7. Compressive strength

The compressive strength was carried out on 50-mm cubes at the ages of 1, 7, and 28 days in accordance with ASTM C109 [59]. All specimens were cured at room temperature ( $23 \pm 2^\circ \text{C}$ ) and sealed with a plastic sheet to prevent any moisture loss until the moment of testing.

### 3. Results and analysis

#### 3.1. Characterisation of RHA-derived sodium silicate and residues

The characterisation of RHA-derived sodium silicate was carried out using ICP-OES, XRD, XRF, TG and FTIR analyses to assess the efficiency of the extraction of silica from RHA.

##### 3.1.1. Dissolution efficiency

After the dissolution process, the solution was filtered to quantify the amounts of residues and dissolved oxides. After the filtration process, the weight of undissolved particles (residues) was  $\sim 115 \text{ g}$  of the suspension, which indicates that  $\sim 371 \text{ g}$  of RHA was dissolved from the  $486 \text{ g}$  that was used in the suspension. Thus, the dissolution efficiency of RHA was calculated to be 76.3%. The remaining 23.7% portion seems to be composed of undissolved silica, unburnt carbon, and some minor crystalline oxides. Moreover, ICP-OES analysis was carried out on the filtered solution to measure the Si and Na concentrations. The ICP-OES results revealed that the weight percentages of dissolved oxides in the RHA-derived solution were 25.7%  $\text{SiO}_2$ , 16.0%  $\text{Na}_2\text{O}$  and 58.3%  $\text{H}_2\text{O}$  with a  $\text{SiO}_2/\text{Na}_2\text{O}$  weight ratio of 1.6. The  $\text{SiO}_2/\text{Na}_2\text{O}$  ratio was lower than the targeted ratio of 2.0 found in the commercial SS solution ( $\text{Na}_2\text{O}$

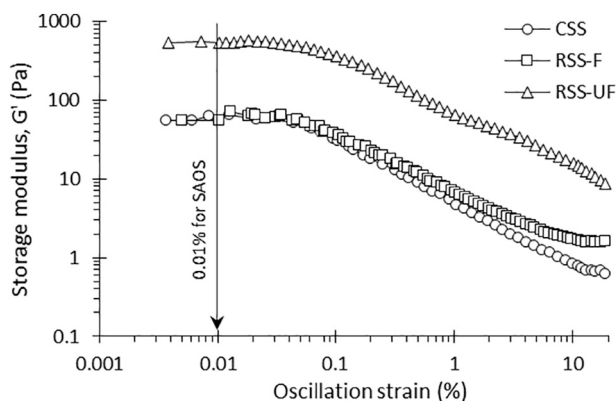


Fig. 2. Oscillation strain sweep of AAM pastes made of commercial and RHA-derived SS.

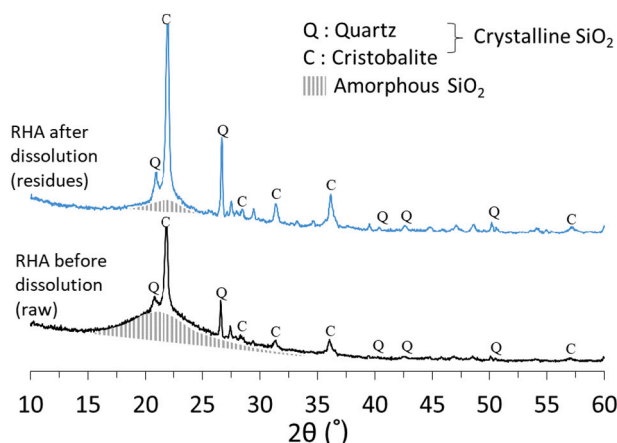


Fig. 4. XRD pattern of RHA before (raw) and after (residues) dissolution.

= 14.7%,  $\text{SiO}_2$  = 29.4%, water = 55.9% in the commercial SS) because RHA was not completely dissolved.

### 3.1.2. XRF and XRD analyses

The XRD patterns in Fig. 4 showed that the raw RHA was mostly amorphous with the presence of quartz and cristobalite phases. The XRF analysis showed that the raw RHA contained 91.9%  $\text{SiO}_2$ , 6.1% other oxides and 2% unburnt carbon. From the Rietveld XRD quantification refinement, the raw RHA was composed of 85.5% of amorphous phase and 14.5% of crystalline phases (quartz and cristobalite). Assuming that the amorphous phase mainly was non-crystalline  $\text{SiO}_2$ , 91.9% of  $\text{SiO}_2$  in the raw RHA (obtained from the XRF analysis) can be segmented as 78.6% and 13.3% of amorphous and crystalline silica, respectively.

After the dissolution, the amorphous phase of silica represented by a broad hump between  $15^\circ$  and  $30^\circ$   $2\theta$  was clearly diminished in conjunction with strong peaks of quartz and cristobalite. The residues (the remaining 23.7% undissolved particles) obtained from the filtration process were characterised. The XRF analysis showed that the residues contained 76.4%  $\text{SiO}_2$ , 14.6% other oxides and 9% unburnt carbon. From the Rietveld XRD analysis, the residues contained 48% of amorphous  $\text{SiO}_2$  and 52% of crystalline phases. Therefore, 76.4% of total  $\text{SiO}_2$  (from XRF) in the residues were composed of 36.7% and 39.7% of amorphous and crystalline silica, respectively.

### 3.1.3. Thermogravimetric analysis (TGA)

The results of TGA on RHA powders before (raw RHA) and after the hydrothermal process (residues) are shown in Fig. 5. The raw RHA showed a final weight loss of about 2.5%. The weight loss in the RHA

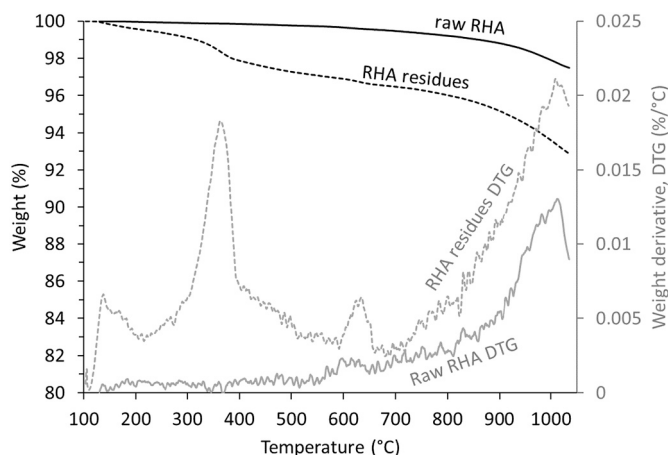


Fig. 5. TGA of RHA before (raw) and after (residues) dissolution.

would be the result of the loss of the inorganic carbon and volatile matters during heating in the inert gas (nitrogen gas) [60]. As shown in Fig. 5, the weight loss of the residual RHA ( $\sim 7.1\%$ ) was much higher than the raw RHA. This is well expected because the relative carbon content and volatiles in the residues would increase by dissolving the soluble silica during the hydrothermal process [25]. In addition, the pores in the residual particles may still have some of water even after the filtration and drying. This water could contribute to the weight loss at  $\sim 150^\circ\text{C}$  in Fig. 5. Another distinct weight loss was observed at  $\sim 350^\circ\text{C}$  in the residues. This seems to be the result of the decomposition of the SS gel that remained in the pores of the residual particles. Similar weight loss was observed by Rao et al. [61] at  $325^\circ\text{C}$ . Moreover, the release of the volatile products generated during the decomposition of biomass also contributes to the weight loss at  $\sim 350^\circ\text{C}$  [62].

### 3.1.4. FTIR analysis

Fig. 6 presents the FTIR spectra of the raw RHA and the residual RHA. The main bands assigned to the Si—O—Si bonds in the raw RHA and the undissolved particles are observed in the spectrum between  $700$  and  $1300\text{ cm}^{-1}$ . The FTIR absorbance of RHA shows a significant decrease in the intensity of all bands after dissolution, suggesting the efficiency of the dissolution process. Still, traces of undissolved crystalline silica are observed through the Si—O—Si groups at  $1051\text{ cm}^{-1}$  and  $793\text{ cm}^{-1}$  in the residues [26]. FTIR spectra for three sodium silicate solutions (commercial SS, filtered RHA SS and unfiltered RHA SS) were also obtained to understand the spectral changes of the Si—O—Si bonds. Two peak positions were observed in commercial and unfiltered RHA SS solutions at  $984\text{ cm}^{-1}$  and  $860\text{ cm}^{-1}$ , along with a shoulder at  $\sim 1100\text{ cm}^{-1}$ , as shown in Fig. 6. On the other hand, filtered RHA SS solution showed one peak at  $984\text{ cm}^{-1}$  and a trace of the shoulder at  $\sim 1100\text{ cm}^{-1}$ . These bands are assigned to vibrations of different silicate species [44]. Both commercial and RHA-derived SS contained a large portion of monomers represented by a high intensity of absorption at  $984\text{ cm}^{-1}$ . However, the commercial SS showed a more prominent shoulder at  $\sim 1100\text{ cm}^{-1}$  compared to the filtered RHA SS, which indicates the presence of more condensed silicate species [63]. The silicate structure primarily depends on the  $\text{SiO}_2/\text{Na}_2\text{O}$  ratio [63]. As a result, the relative intensity of the condensed silicate species at  $\sim 1100\text{ cm}^{-1}$  is more

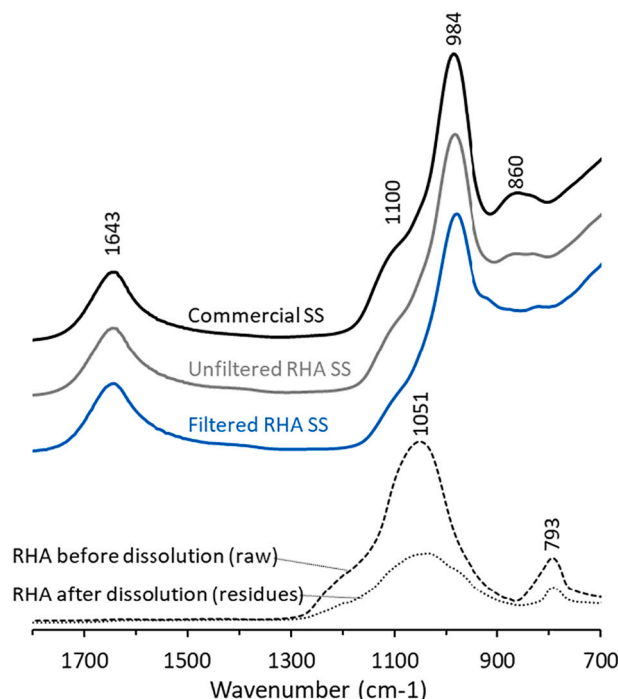


Fig. 6. FTIR spectra of RHA, undissolved particles, and SS solutions.

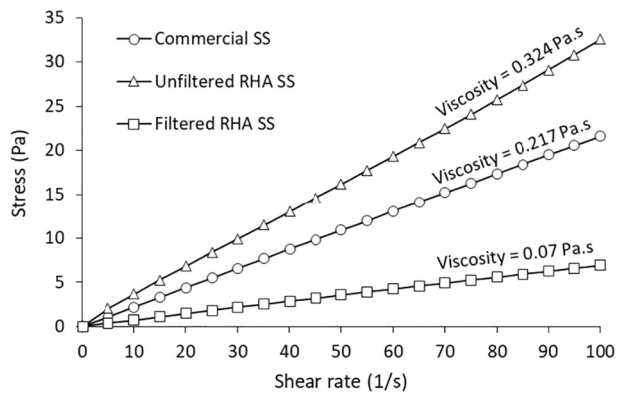


Fig. 7. Flow curves of commercial SS and RHA-derived SS solutions.

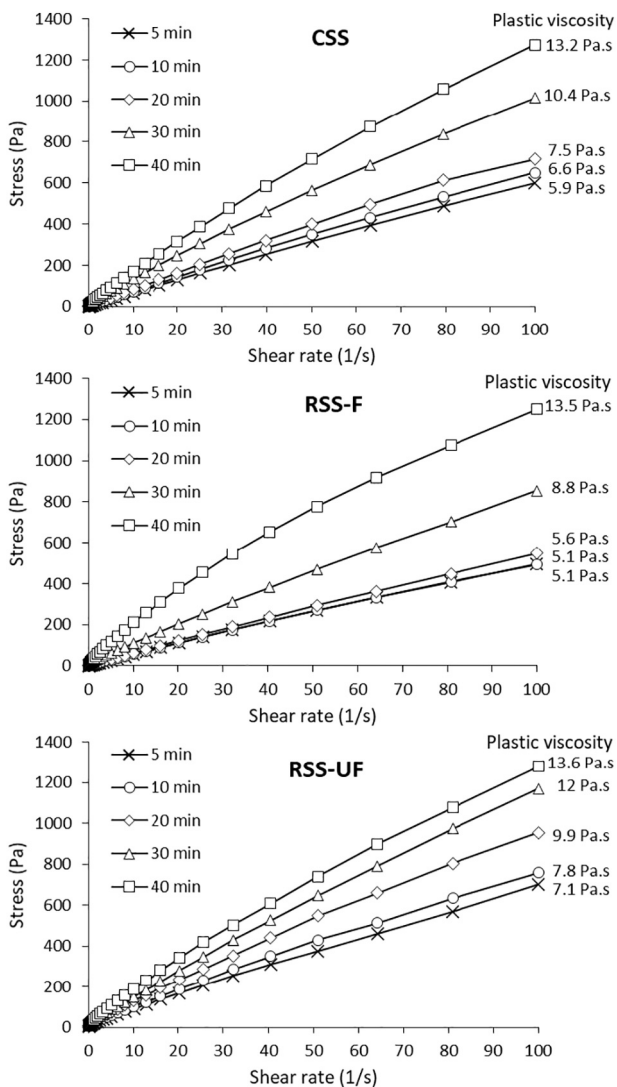


Fig. 8. Flow sweep curves of AAM pastes made of (a) commercial SS, (b) filtered RHA SS and (c) unfiltered RHA SS at different elapsed times.

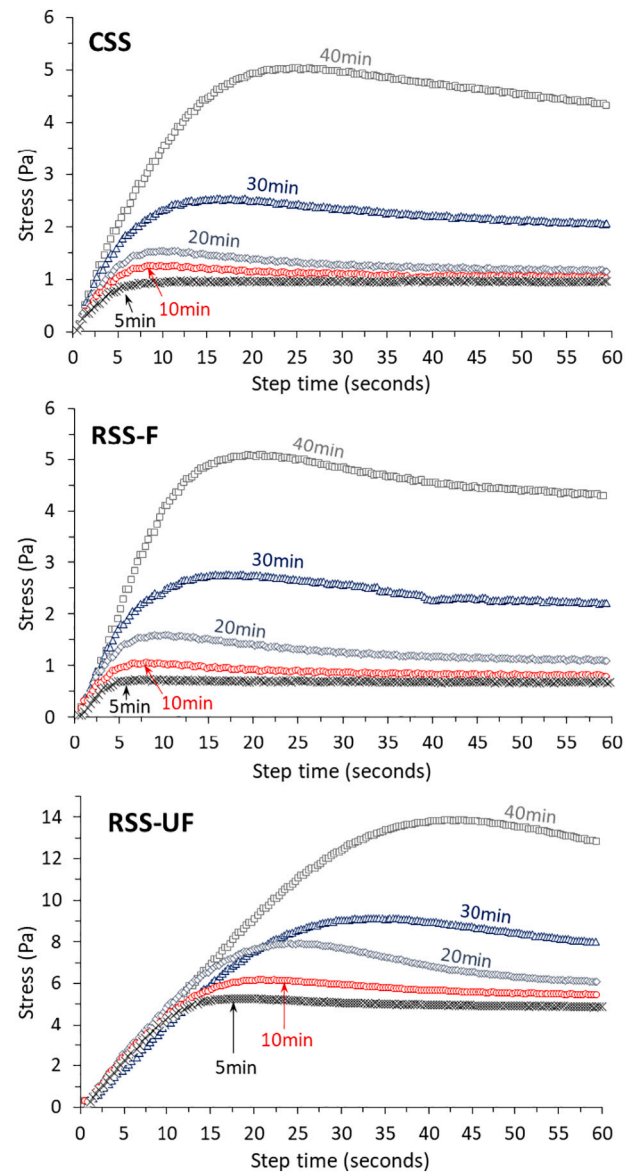


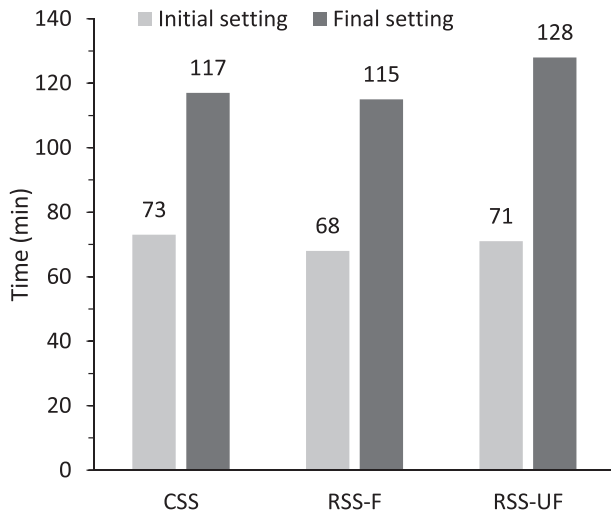
Fig. 9. Static yield stress of AAM pastes made of commercial SS, filtered RHA SS and unfiltered RHA SS at elapsed times.

prominent in the commercial SS due to the relatively higher availability of silicates and the higher  $\text{SiO}_2/\text{Na}_2\text{O}$  ratio in this solution.

### 3.1.5. Newtonian viscosity of the activators

The viscosities of the SS solutions were determined using the flow sweep test. As shown in Fig. 7, the results show that all activators were Newtonian fluids. The commercial SS had a viscosity value ( $0.217 \text{ Pa}\cdot\text{s}$ ) higher than the one from the filtered RHA SS ( $0.070 \text{ Pa}\cdot\text{s}$ ), which seems to be the result of the higher contents of condensed silicate species observed in FTIR spectra. In addition, the impact of the undissolved residues on the flow properties of the RHA-derived SS was clearly observed, showing the highest viscosity of the unfiltered solution ( $0.324 \text{ Pa}\cdot\text{s}$ ).





**Fig. 10.** Initial and final setting times of AAM pastes made of RHA-derived and commercial activators.

### 3.2. Evolution of plastic viscosity, dynamic and static yield stresses of AAM pastes

Fig. 8 shows the evolution of the flow sweep curves of AAM pastes, and Fig. 9 presents the results of the static yield stress measurement of corresponding AAM pastes (made of commercial SS, filtered RHA SS, and unfiltered RHA SS). As shown in Fig. 8, every flow sweep curve shows a good linear behaviour (almost Newtonian fluids). The results of the CSS mix showed an increase in the plastic viscosity from 5.9 Pa·s to 13.2 Pa·s when the resting time was increased from 5 min to 40 min. A similar trend was also observed for the RSS-F mix. The effect of the residues on the plastic viscosity was also examined on the mix RSS-UF. As can be seen in Fig. 8, the plastic viscosity of RSS-UF was generally higher than CSS and RSS-F by up to 40% at 5 min, and this gap was gradually reduced, reaching similar values for all mixes at 40 min.

In the static yield stress measurement (see Fig. 9), the stress resulting from the continuous shear was increased and reached a critical peak

value at which the sample starts to flow. The critical peak stress and the time needed to reach this value were dependent on both the age of the sample and the type of SS used. For instance, the time needed to reach the peak value at the age of 40 min was four times higher than that at the age of 5 min for all AAM pastes. The evolution of the static yield stress overtime for the mixes CSS and RSS-F was similar, as shown in Fig. 9. However, the time needed to reach the peak stress value for the AAM pastes using RSS-UF was longer than the times obtained from the other two mixes (CSS and RSS-F). The static yield stress for the RSS-UF paste was much higher than those obtained from the other two AAM pastes. More details about the comparison between the static and dynamic yield stresses are discussed in Section 4.2.

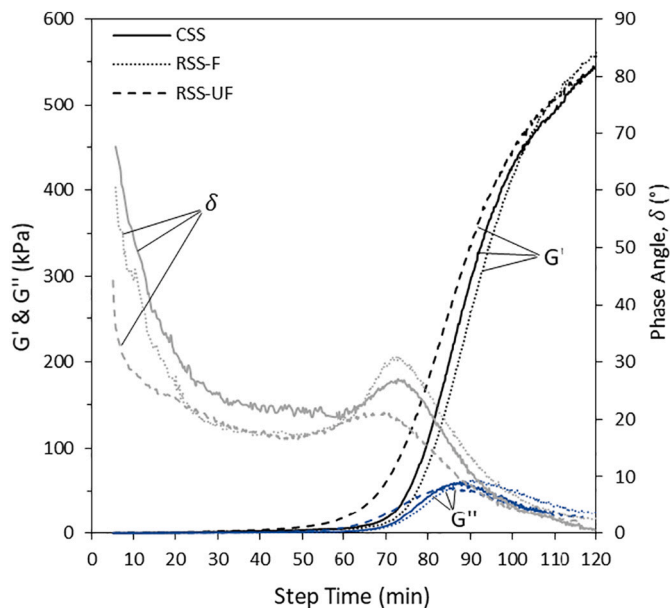
### 3.3. Setting time

Fig. 10 shows the initial and final setting times of AAM pastes made of RHA-derived and commercial activators. The initial setting time of all pastes was varied within a narrow range between 68 min and 73 min. On the other hand, the AAM mix made of unfiltered RHA SS (RSS-UF) had a little longer final setting time (128 min) than those values from the other two pastes (CSS and RSS-F).

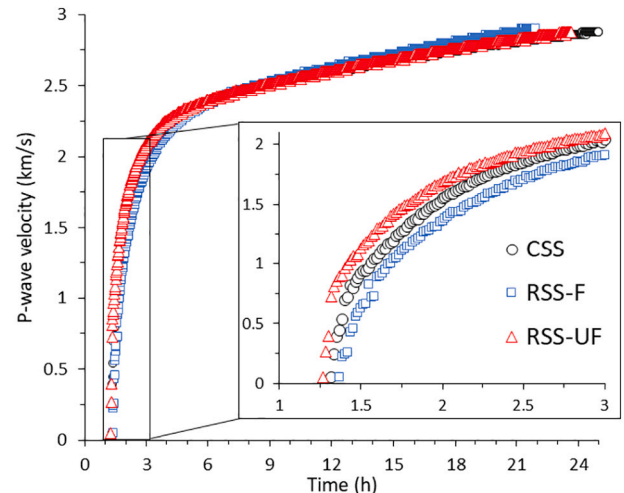
### 3.4. Viscoelastic behaviour

Fig. 11 presents the effect of filtered and unfiltered RHA SS on the temporal changes in the viscoelastic parameters ( $G'$ ,  $G''$  and  $\delta$ ) of AAM pastes in comparison with AAM paste made of commercial SS. The first contact between the activator and precursors was considered as zero time. The temporal changes in  $G'$ ,  $G''$  and  $\delta$  in AAM pastes were thoroughly investigated in the recent literature [35,37,64,65], and the current results in this paper well agreed with the previous results. Alnahhal et al. [35] suggested four distinctive stages during the short-term evolution of the AAM paste - Stages I and II: the liquid-like behaviour and the formation of the weak elastic network, Stage III: transient period, and Stage IV: rigidification stage (solid-like interactions of the formed network due to the chemical reactions).

To facilitate the interpretation of the small changes in  $G'$  and  $G''$  values during the very early Stage I and Stage II ( $< \sim 50$  min), the phase angle,  $\delta$ , that equals to  $\tan^{-1}(G''/G')$  is plotted as a function of time (see Fig. 11). For CSS and RSS-F, the changes in  $\delta$  were more or less identical, and the initial  $\delta$  values started at  $>45^\circ$  ( $G'' > G'$ ), indicating a liquid-like behaviour of the paste at the initial period up to around 15 min. Then, the values of these phase angles decreased to less than  $45^\circ$  because  $G'$  was higher than  $G''$ . On the other hand,  $\delta$  was started below  $45^\circ$  for the mix RSS-UF, suggesting that  $G'$  has quickly exceeded  $G''$ .



**Fig. 11.** Evolution of the viscoelastic parameters as a function of time for AAM pastes made of commercial SS (CSS), filtered RHA SS (RSS-F) and unfiltered RHA SS (RSS-UF).



**Fig. 12.** Evolution of P-wave velocity as a function of time.

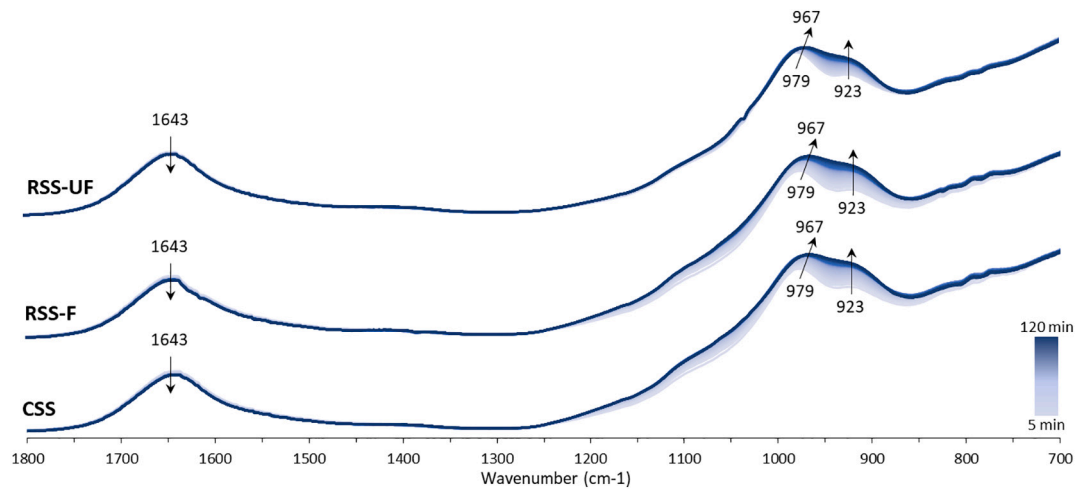


Fig. 13. FTIR spectra of AAM pastes made of commercial and RHA-derived SS.

Between 5 min and 60 min (Stage II), both values of  $G'$  and  $G''$  moderately increased. However, the increasing rate of  $G'$  is greater than that of  $G''$ , and thus the phase angle was continuously decreasing at a moderate rate. Fig. 11 shows that a transient period (Stage III) begins at  $\sim 50$  min for the RSS-UF mix and at  $\sim 60$  min for both CSS and RSS-F mixes. At the transient period,  $\delta$  started to increase. This increase in  $\delta$  means that the  $G''$  evolution was slightly accelerated compared to  $G'$ . This acceleration was diminished at  $\sim 70$  min. It should be noted that at the end of the transient period (i.e.  $\sim 70$  min), the samples CSS, RSS-F and RSS-UF still show relatively very low  $G'$  values of  $\sim 15$  kPa,  $\sim 20$  kPa and  $\sim 60$  kPa, respectively, indicating the weak interactions between particles [35].

The chemical rigidification stage (Stage IV) is defined as the process of forming a stiff network due to chemical reactions (geopolymerisation) [35,66]. In this Stage IV, the elastic portion of the shearing modulus,  $G'$ , was increased drastically to  $\sim 550$  kPa, which implies the transition of the paste to a solid-like state. The onset of rigidification can be found at the peak point of  $\delta$  [35]. The onset of rigidification occurred at 73 min, 71 min and 65 min for CSS, RSS-F and RSS-UF, respectively. In summary, the overall trend of the viscoelastic evolutions obtained from three AAM pastes was quite similar, but the RSS-UF showed a slightly faster evolution of  $G'$  and  $G''$  than the others. The onset of rigidification also agreed well with the initial setting time determined by the Vicat needle test reported in Section 3.3.

### 3.5. Ultrasonic P-wave velocity

Changes in the velocity of the P-waves during the first 24 h for the AAM pastes are shown in Fig. 12. All pastes showed a similar trend and magnitude of the velocity over time. This evolution can be divided into three stages. Prior to the transmission of the waves through the paste, the first stage showed a delay time of 80 min, 82 min and 77 min for the mixes CSS, RSS-F and RSS-UF, respectively, suggesting the liquid-like behaviour of the sample. It is interesting to note that the timing of the first measurement of P-wave velocity well agreed with the initial setting time (Section 3.3) and the onset of the rigidification (Stage IV) obtained from the SAOS test (Section 3.4). This correlation will be discussed further in Section 4.3. In the second stage, the P-waves showed a sharp increase from  $\sim 0.25$  km/s to  $\sim 2$  km/s during the first 3 h. A steady increase in the P-wave velocity was observed in the third stage, reaching a threshold value of  $\sim 3$  km/s after 24 h.

### 3.6. Gel formation in the early stages of reaction

In-situ ATR-FTIR spectroscopy was used to study the effect of RHA-derived activator on the mechanism of AAM gel formation. Fig. 13 illustrates the changes in FTIR spectra in the first 120 min of reaction. Faded colours show the initial FTIR spectra collected every minute, and the direction of changes for the main bands is represented by arrows. As

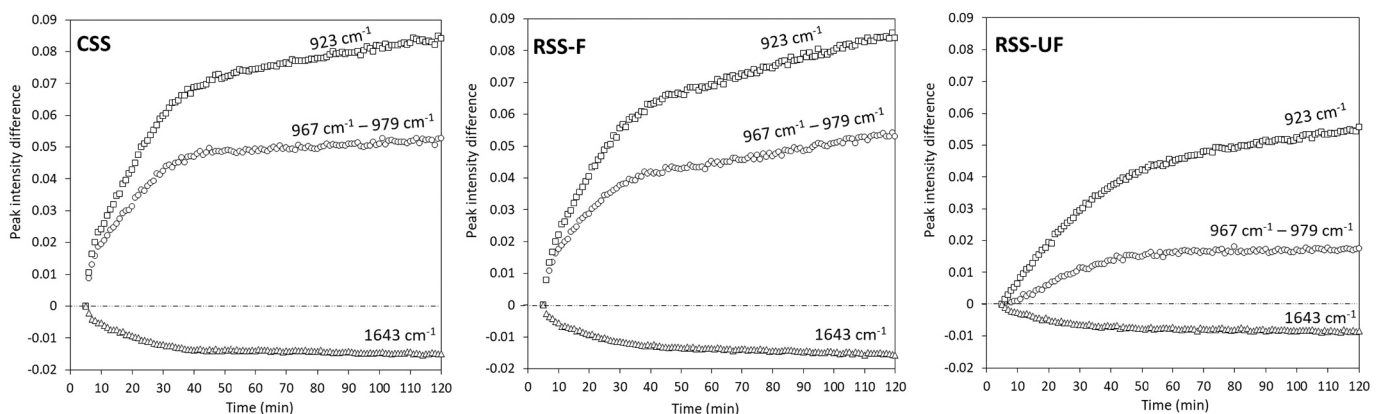


Fig. 14. Changes in the intensity of FTIR bands over time. Peak intensity difference at time “x” = peak intensity at time “x” – peak intensity at 5 min.



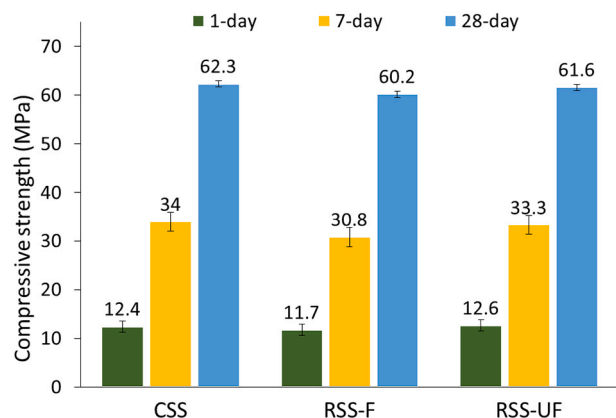


Fig. 15. Compressive strength of AAM pastes made of commercial and RHA-derived SS.

can be seen, all pastes showed similar initial band peaks: one peak at  $1643\text{ cm}^{-1}$  for bending vibration of water and two peaks at  $923\text{ cm}^{-1}$  and  $979\text{ cm}^{-1}$  representing the vibrations of the Si—O—(Si, Al) bonds in the growing gels [58]. Moreover, immediately after mixing, the intensity at  $979\text{ cm}^{-1}$  was higher than  $923\text{ cm}^{-1}$  due to the presence of silicate species in the activating solution. Fig. 14 illustrates the three absorbance intensities for three peaks ( $1643\text{ cm}^{-1}$ ,  $923\text{ cm}^{-1}$ , and  $979\text{--}967\text{ cm}^{-1}$ ) as a function of time. Over time, the intensity at  $1643\text{ cm}^{-1}$  decreases, suggesting the consumption of water in the reaction [67]. The increased intensity at  $923\text{ cm}^{-1}$  and  $979\text{ cm}^{-1}$  can be seen as an indication of the growing gel network [68]. The position of the band at  $979\text{ cm}^{-1}$  was shifted to  $967\text{ cm}^{-1}$  within the first few hours of reaction. This shift was resulted from the structural change due to the development of the silicate network [68]. This band is expected to overlap with the band at  $923\text{ cm}^{-1}$  at longer reaction times to form one peak at  $\sim 960\text{ cm}^{-1}$  [69]. In general, in the first 50 min, all pastes showed a sharp increase in the intensity of the band at  $923\text{ cm}^{-1}$  and  $967\text{--}979\text{ cm}^{-1}$ , coinciding with a drop in the intensity of  $1643\text{ cm}^{-1}$ . After that, the intensities of the bands gradually increased at lower rates.

### 3.7. Compressive strength

The compressive strength of AAM pastes made of commercial SS, and two RHA-derived SSs is shown in Fig. 15. Generally, the compressive strength was comparable in all mixtures. This confirms the equivalence in the mix designs between all mixes, where similar availability of alkalis and silicate species were added.

## 4. Discussion

### 4.1. Extraction efficiency and quality of sodium silicate from RHA

An alternative SS was produced from RHA using heat treatment with NaOH (stirring for 1 h at  $80\text{ }^{\circ}\text{C}$ ). The summary of the quantification analysis of the raw RHA and the residues after the hydrothermal process was determined based on XRF, Rietveld XRD and TG analyses are plotted in Fig. 16. The dissolution efficiency computed using the weight of the residue was 76.3% (see Section 3.1). A previous study also showed that the yield of silica is optimum at  $80\text{ }^{\circ}\text{C}$ , and no further significant dissolution was recorded at higher processing temperatures [20]. Dissolution of silica minerals is influenced by many factors, including the type of silica phase, the surface area of silica mineral, reaction temperature, pH value of the solution, type and concentration of ionic species present in the solution, and the degree of saturation with respect to the dissolving component [70]. The raw RHA contained 78.6% of amorphous  $\text{SiO}_2$  that had a much higher dissolution rate than other crystalline  $\text{SiO}_2$  minerals. The solution used to dissolve the raw RHA was

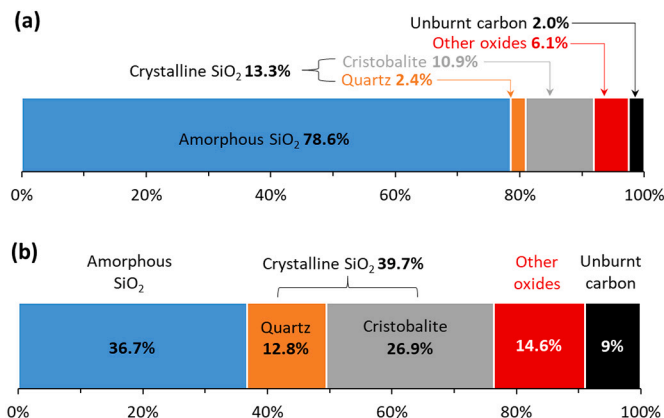


Fig. 16. Quantification of RHA phases (a) before and (b) after the dissolution process based on XRF, Rietveld XRD and TG analyses.

7.0 mol/l of NaOH solution ( $\text{pH} \sim 14$ ), of which the condition accelerated the dissolution of silica. Therefore, the amorphous nature of the RHA and the high alkalinity of the NaOH solution contributed to achieving high dissolution rate of 76.3%.

### 4.2. Flowability

The evolution of the flow properties of AAM pastes made of filtered/unfiltered RHA SS and commercial SS was determined as a function of time. Fig. 17 compares the temporal evolution of both dynamic and static yield stresses obtained from three AAM pastes. The undissolved particles from the unfiltered RHA SS in the mix RSS-UF increased the yield stress and plastic viscosity. The unburnt carbon in the residues could be an important factor that resulted in a low flowability of RSS-UF compared to CSS and RSS-F. The presence of unburnt carbon in the system affects the rheological properties due to the increased water demand [71,72]. This was evident in the mix RSS-UF compared to CSS and RSS-F, where the unfiltered solution used in RSS-UF contained residues that were composed of 9% of unburnt carbon. This higher yield stress obtained from the RSS-UF than those from the other two pastes seems to cause the longer time needed to reach the peak stress in the stress growth test, as shown in Fig. 9. Stiffening due to the chemical reaction would be more accelerated when the pastes are close to the initial setting, and thus this trend can be clearly observed in the later age shown in Fig. 17.

### 4.3. Structural build-up

The availability of silicate species in the alkaline activator is a key component that affects the conversion of solid precursors into a geopolymeric gel (reaction products) [68] and the consequent temporal chemical and physical changes in AAM systems (initial and final settings) [73,74]. This paper intended to develop the RHA-derived SS comparable to the commercial SS. Thus, the flowability and the temporal viscoelastic changes caused by geopolymerization was expected to be similar for three AAM pastes prepared. These results are in line with previous studies on the evolution of the flow properties and viscoelastic parameters of AAM made of different amounts [34] and sources [75] of silica. Poulesquen et al. [75] found major differences in the geopolymerization kinetics of AAM synthesised from two types of hydroxides (sodium and potassium hydroxides), but minor differences when two types of silica (fused and precipitated silica) were used in the activator.

To facilitate the understanding of the comprehensive correlation among the different time-dependent tests performed in this paper, the SAOS, ultrasonic P-wave velocity, FTIR and setting time tests were combined in Fig. 18 for a representative mix (CSS). The SAOS test

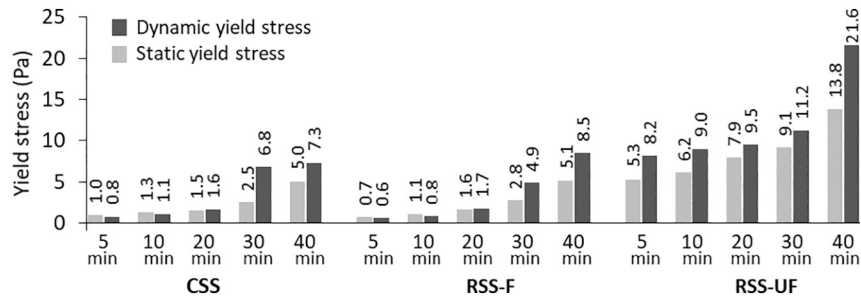


Fig. 17. Comparison between dynamic and static yield stress of AAM paste made of commercial and RHA-derived SS.

enabled us to characterise the interparticle interactions by allowing the particles to recover the small deformation elastically so that  $G'$ ,  $G''$  and  $\delta$  can be monitored over time. Low values of  $G'$  and  $G''$  indicate the non-percolated system until the initial setting [35]. The magnitude of  $G'$  for the mixes CSS and RSS-F was  $\sim 20$  kPa before the initial setting. This indicates the formation of a weakly elastic non-percolated network. The studies by Alnahhal et al. [35] and Rouyer and Poulesquen [37] concluded that up to the initial setting, the negligible colloidal interactions between particles result in the non-percolated system. This is a distinctive difference compared to the Portland cement system in which a well-percolated system is developed within a few seconds after mixing [76]. The very low yield stresses obtained from flow sweep and stress growth tests (see Fig. 17) also support the presence of a non-percolated network in the mixes CSS and RSS-F.

It should be noted that P-wave velocity strongly supports this non-percolated system because the first P-wave velocity was recorded just after the initial setting time from the Vicat test (see Fig. 18). The P-wave velocity indicates the connectivity and total solid phase in the microstructure [40]. Numerous studies on cement paste [77–79] detected a slow propagation of P-waves through the paste at the very fresh condition due to the particle connection (percolation), and then the P-wave velocity was accelerated at the hardened state. However, no P-wave velocity was detected from all three AAM pastes used in this study until the initial setting. This absence of P-wave velocity indicates the non-percolated system of AAM pastes. However, it should be noted that this P-wave velocity may be detected at the fresh condition if solid content increases, and thus this will be affected by the several parameters in the mixture design, including the viscosity of activator, liquid to solid volume ratio, and type of precursors.

After  $\sim 70$  min, a solid percolation starts to form due to the progressive formation of interconnected reaction products. Indeed, as illustrated in Fig. 18, the sharp increase in  $G'$  and P-wave velocity and the subsequent low values of  $\delta$  well reflect this accelerated reaction. It is

interesting to note that the Vicat initial setting time (73 min) well coincided with i) the onset of the rigidification time (73 min) determined from SAOS, and ii) the time at the first record of the P-wave velocity (78 min) (see Fig. 18). The other two AAM pastes (RSS-F and RSS-UF) also showed the same trend. These combined results revealed that the percolation path of interconnected clusters in AAM systems that characterises the transition from liquid-like to solid-like state occurs only after stiffening. This late formation of the percolation path is distinctly different from the cement paste system that has the percolation path within a few minutes after mixing [35].

AAM pastes activated with commercial and RHA-derived SS showed a similar position of the Si—O—(Si or Al) bands at  $923\text{ cm}^{-1}$  and  $979\text{ cm}^{-1}$ . Then, the positions of the bands for three AAM pastes at  $979\text{ cm}^{-1}$  was shifted to the lower wavenumber of  $967\text{ cm}^{-1}$ , which suggested that reaction products from RSS-F and RSS-UF had a comparable structure to the CSS (incorporated with the commercial SS). This also proved the similarity of silicates in the derived activator. Moreover, Fig. 14 shows a relatively sharp increase in the intensity of the bands at  $923\text{ cm}^{-1}$  and  $967\text{ cm}^{-1}$  in the period between 5 min and  $\sim 50$  min. This increase is attributed to the dissolution of precursors in the presence of silicate species, which in turn facilitates the formation of aluminosilicate oligomers and gels [80]. The overall trend in all pastes was similar.

However, in Fig. 15, RSS-UF showed lower absorbance intensity compared to CSS and RSS-F. This indicated that RSS-UF had a slower dissolution of precursors and, consequently, slower geopolymerisation than those from CSS and RSS-F. This could be attributed to the presence of a relatively high amount of unburnt carbon in the RSS-UF mix. A previous study pointed out that the presence of high carbon in the system results in a lower dissolution rate [81]. A longer final setting time shown in Fig. 11 may be the result of this slower reaction in RSS-UF. On the contrary, for RSS-UF, the onset of the rigidification (65 min) and the time for the first record of the P-wave velocity (77 min) were slightly earlier than those times obtained from CSS and RSS-F (see Figs. 12 and

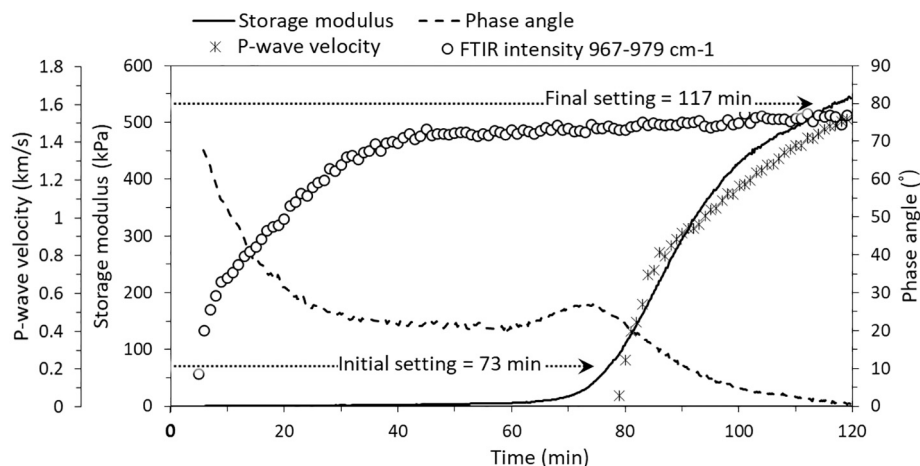


Fig. 18. The evolution of storage modulus ( $G'$ ), phase angle  $\delta$ , ultrasonic P-wave velocity, and FTIR intensity at  $967\text{--}979\text{ cm}^{-1}$  for the mix CSS.

13). This seems to be caused by the presence of undissolved particles in the activator that act like a filler, which in turn increases the particle packing. Therefore, the physical interaction of the residual particles in RSS-UF may compensate for the slightly delayed reaction. Thus, all three pastes had a similar mechanical strength at later ages (see Fig. 16). Studies on the durability and mechanical performance of AAM using filtered and unfiltered activators would be required to determine the effect of the residual particles in the unfiltered activators.

Overall, despite the minor difference in the structural build-up between the pastes without the residues (CSS and RSS-F) and the paste with residues (RSS-UF), the results presented in this paper are practically comparable. In addition, the similar availability of silicates in the activator can be achieved using the filtered and unfiltered RHA SS. Thus, using the unfiltered RHA SS can be more efficient due to eliminating the filtration process that consumes time and energy. Nonetheless, more research is needed on durability and mechanical performance.

## 5. Conclusions

A series of tests were conducted to assess the suitability of using an alternative SS derived from RHA in AAMs. The effect of filtered and unfiltered RHA SS on the structural build-up of AAM pastes was evaluated in terms of rheology, P-wave velocity, setting time and FTIR spectra, which provided insights into the flowability, viscoelastic evolution and gel formation mechanism over time. The following conclusions can be drawn:

- The results revealed that the reactive (amorphous) silica in RHA plays an essential role in controlling the amount of silicate species in the extracted SS. The amorphous silica was almost completely dissolved in the NaOH solution after stirring for 1 h at 80 °C. Quartz, cristobalite, and unburnt carbon were detected in the RHA residues after dissolution, indicating the insolubility of these phases into the alkaline solution.
- Although RHA SS contained a larger portion of monomers and less condensed silicate species than commercial sodium silicate, the overall structural build-up of AAM paste activated using filtered RHA SS were very comparable to the one activated using commercially available SS. This suggests that the formation of identical reaction products and the minor effect of the nature of the silicate species in the activator.
- Using unfiltered RHA SS containing undissolved particles is still practically viable, but it increases the yield stress. In addition, a slight delay in the final setting and the gel formation was observed in the mix made of the unfiltered RHA SS. However, all three pastes showed a very similar compressive strength development for 28 days.
- The evolution of the viscoelastic parameters, P-wave velocity and setting time indicate that the network formation occurs in two phases: A weakly elastic non-percolated network is formed in the first phase before the initial setting followed by the formation of a solid percolation path due to stiffening in the second phase. The initial weak non-percolated network was formed due to the use of SS of relatively high viscosity. This resulted in delayed flocculation represented by a delay time before the transmission of the P-waves through the sample. The evolution was similar in all mixes, showing that the mechanisms involved are similar regardless of the nature of the SS used.

Although RHA-derived sodium silicates offer many advantages, these activators still require further research to be successfully implemented in AAMs. Future research on the effect of various processing parameters on the availability and structure of silicate species during the preparation of the RHA-derived sodium silicates is highly recommended.

## Author statement

All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript.

## CRediT authorship contribution statement

**Mohammed Fouad Alnahhal:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Abdelrahman Hamdan:** Formal analysis, Investigation, Writing – review & editing. **Ailar Hajimohammadi:** Conceptualization, Methodology, Validation, Resources, Writing – review & editing, Supervision. **Taehwan Kim:** Conceptualization, Methodology, Validation, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

## Declaration of competing interest

We confirm that there is no conflict of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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